POKHARA UNIVERSITY

Level: Bachelor
Programme: BE
Course: Discrete Structure

Semester: Fall

Year : 2024
Full Marks : 100
Pass Marks : 45
Time : 3 hrs.

Candidates are required to give their answers in their own words as far as practicable.

The figures in the margin indicate full marks.

Attempt all the questions.

1. a) Define Bijective function. Express GCD of 161 and 28, and express it as a linear combination of 161 and 28. 8

   b) What are Permutation and combination? Explain with example. 7

2. a) Let A = Z, the set of integers and let, R = {(a, b): a≤b}. Is R reflexive, symmetric, anti-symmetric or transitive? 7

   OR

   Let A= {w, x, y, z} and R = {(w, x), (x, w), (x, y), (y, z), (z, w)}. Find the transitive closure of R.

   b) Define equivalence relation. Is the relation, {(0,0), (1,1), (2,2), (3,3)} on {0, 1, 2, 3} an equivalence relation? Justify your answer. 8

3. a) Solve the following recurrence relation. 8

   $a_n = 3a_{n-1} + 2n$ where $a_1=3$

   b) Find the solution of recurrence relation: 7

   $a_n = 6a_{n-1} - 11a_{n-2} + 6a_{n-3}$ with $a_0 = 2$, $a_1 = 5$ and $a_2 = 15$

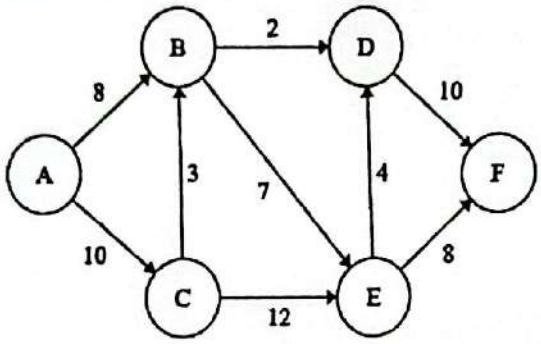
4. a) Using proof by contrapositive show that if $n^3 + 2n + 1$ is odd then n is even. 7

   b) Define logical equivalence with reference to De Morgan's Law. State the converse, contrapositive, and inverse of the conditional statement: "*If it snows tonight, then I will stay at home.*" 8

   OR

   Explain different quantifiers used in predicate logic with suitable example. Is $\forall_x \exists_y (P(x,y))$ and $\exists_y \forall_x (P(x,y))$ equivalent?

5. a) Find the maximum flow in the given network. 7



b) Differentiate between Hamilton graph and Euler's graph along with necessary diagrams. 8

6. a) Explain Chomsky Hierarchy of Grammar in detail. 8

b) Design a DFA that accepts any string over$\{a, b\}$ which doesn't contain "aabb" string. 7

7. Write short notes on: (**Any two**) 2×5

a) Application of Graph Theory

b) Boolean Matrix operation

c) Connectives of propositional logic